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Paper 1: Physiologically Based Pharmacokinetic Model of Brigatinib in Healthy Volunteers

Bibliographic Information

- **Title:** Physiologically Based Pharmacokinetic Model of Brigatinib in Healthy Volunteers and Patients With Cancer
- **Authors:** Xu Y, Huang X, Zhang P, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026;15(8)
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Paper Type: POPK_PKPD

Quick Take

This study develops a whole-body PBPK model for brigatinib in PK-Sim, validated against European clinical data, then extrapolates to Chinese populations and special populations (hepatic/renal impairment, DDIs). The model predicts that while severe hepatic (CP-C) and renal impairment increase exposure (137% and 177% of healthy, respectively), mild-to-moderate hepatic impairment does not increase total exposure. Clinicians should consider dose adjustments for Chinese patients with severe organ impairment or those on strong CYP3A4 modulators.

Executive Summary

This paper presents a comprehensive PBPK modeling framework for brigatinib, an ALK inhibitor used in metastatic NSCLC, with the goal of informing dosing in Chinese patients and special populations. Using PK-Sim, the authors developed and validated a whole-body PBPK model against European clinical data from food effect, DDI, and organ impairment studies, achieving good predictive performance (GMFE < 1.53). The model was then extrapolated to Chinese populations, predicting that mild-to-moderate hepatic impairment does not increase brigatinib exposure, while severe hepatic and renal impairment increase exposure by ~37% and ~77%, respectively. The model also quantified DDI magnitude (2-fold increase with itraconazole, 80% decrease with rifampin) and provided dose adjustment recommendations for Chinese patients. The work demonstrates the utility of PBPK modeling for bridging across populations and special conditions, though the lack of Chinese clinical PK data for direct validation remains a limitation.

Scientific Context & Motivation

Brigatinib, a second-generation ALK inhibitor, is approved for ALK-positive NSCLC but lacks specific dosing guidance for Chinese patients and those with organ impairment. While the drug is primarily metabolized by CYP3A4, with renal clearance playing a minor role, the impact of hepatic and renal dysfunction on its pharmacokinetics in Asian populations remains poorly characterized. PBPK modeling offers a mechanism-based approach to bridge clinical data from

European populations to Chinese populations and to simulate untested clinical scenarios. This study addresses the gap in personalized dosing recommendations for brigatinib in Chinese patients, particularly those with hepatic or renal impairment, and those on concomitant CYP3A4-modulating medications.

✂ Methodological Snapshot

A whole-body PBPK model was developed using PK-Sim (Open Systems Pharmacology Suite, v11.1) with an 18-compartment structure. The model incorporated physicochemical properties (logP 1.80, fu 0.66, MW 584.1 g/mol), permeability-limited absorption, and CYP3A4/CYP2C8-mediated metabolism. The model was trained on European healthy volunteer data (food effect and DDI studies) and validated against independent datasets including hepatic/renal impairment studies. Virtual populations (n=100, 50% female, age 18-60) were generated using European and Japanese templates (as a surrogate for Chinese). The model was then extrapolated to Chinese populations and used to simulate PK under various conditions, with dose adjustments proposed to match European exposure levels.

☒ Structural Model Breakdown

The PBPK model uses a whole-body structure with 18 compartments representing major organs and tissues (liver, kidney, lung, heart, brain, bone, muscle, skin, spleen, pancreas, gut, adipose, etc.), each with perfusion-limited distribution. The model includes: (1) Absorption: ACAT model with permeability-limited absorption, incorporating specific intestinal permeability ($3.61\text{E-}07$ cm/min) and organ permeability ($4.62\text{E-}05$ cm/min); (2) Distribution: perfusion-limited (well-stirred) model with tissue-to-plasma partition coefficients calculated using the Rodgers and Rowland method; (3) Metabolism: Michaelis-Menten kinetics for CYP3A4 (kcat 1.55 L/min) and CYP2C8 (kcat 4.0 L/min); (4) Elimination: renal clearance of 1.9 L/h. The model was parameterized with physicochemical properties (logP 1.80, fu 0.66, MW 584.1 g/mol, pKa base 8.54, pKa acid 12.88) and validated against clinical data.

Detailed Methodological Analysis

Modeling Approach

Whole-body PBPK model with 18 compartments in PK-Sim v11.1. The model used a bottom-up approach informed by in vitro data and a middle-out approach using structural analogs. Absorption was modeled using an ACAT model with permeability-limited absorption. Metabolism was modeled via CYP3A4 (kcat 1.55 L/min) and CYP2C8 (kcat 4.0 L/min) pathways. The model was developed using fasted-state data from food effect and DDI studies, then validated against fed-state, DDI, and organ impairment data without parameter adjustment.

Data Sources

Retrospective analysis of published clinical data: (1) food effect study (single 180 mg dose, fasted/fed), (2) DDI study (90 mg brigatinib with itraconazole 200 mg; 180 mg with rifampin 600 mg),

(3) hepatic impairment study (Child-Pugh A-C), and (4) renal impairment study (severe, CKD stage 4-5). PK data were extracted from published figures using WebPlotDigitizer v4.2, and non-compartmental analysis was performed using Phoenix WinNonlin v8.3.5.340.

Estimation Methods

Parameter estimation was performed using a combination of literature-derived values and fitting to FDA reports. The model used a stepwise parameter optimization approach. Sensitivity analysis was conducted using change factors of 1.10, 1.05, 1/1.05, and 1/1.10 to assess parameter influence on AUC and Cmax. Virtual populations were generated using PK-Sim's population module with 100 individuals per cohort.

Model Evaluation

Model performance was evaluated by comparing predicted vs. observed plasma concentrations using goodness-of-fit plots (Figure 1) and calculating geometric mean fold error (GMFE) for Cmax and AUC0-∞. Acceptance criteria were set at 0.5-2.0 fold for all predictions. Visual predictive checks were performed by comparing predicted 5th-95th percentile ranges with observed data. The model was validated against independent datasets from hepatic/renal impairment and DDI studies without parameter adjustment.

Covariate Analysis

Covariates were not analyzed using traditional stepwise regression. Instead, the PBPK framework incorporated physiological differences between populations (European vs. Japanese/Chinese templates) and conditions (hepatic/renal impairment) by modifying system-specific parameters such as organ blood flow, plasma protein binding, GFR, and organ volumes. For hepatic impairment, Child-Pugh class-specific pathophysiological changes were incorporated, including reduced plasma protein binding and altered organ blood flow. For renal impairment, the CKD module in PK-Sim was used to modify GFR, renal blood flow, and other relevant parameters.

Statistical Rigor Assessment

The study employs a well-established PBPK modeling framework with appropriate validation strategies. The use of GMFE as a measure of predictive performance is appropriate, with all values falling within the pre-specified 0.5-2.0 fold acceptance range. The model was validated against multiple independent datasets (food effect, DDI, hepatic/renal impairment) without parameter adjustment, strengthening confidence in its predictive ability. However, several limitations exist: (1) PK data were extracted from published figures rather than obtained directly from investigators, introducing potential digitization errors; (2) the sample size for virtual populations (n=100) may not fully capture inter-individual variability; (3) the use of Japanese population as a surrogate for Chinese populations may introduce bias; (4) no formal uncertainty analysis was performed on model parameters; (5) the model's predictions for mild/moderate renal impairment lack clinical validation; and (6) the study did not account for potential differences in drug metabolism enzyme expression between populations.

☒ Key Findings

The PBPK model successfully predicted brigatinib PK across multiple scenarios. In hepatic impairment, the model predicted that AUC_{0-∞} was 89.3%, 99.2%, and 137.4% of healthy values for CP-A, CP-B, and CP-C, respectively, with GMFE ranging from 1.18-1.48 for C_{max} and 1.24-1.53 for AUC_{0-∞}. For severe renal impairment, AUC_{0-∞} was 177.2% of healthy values. DDI simulations showed itraconazole doubled brigatinib AUC_{0-∞}, while rifampin reduced exposure to 20% of control. In Chinese populations, dose adjustments were recommended: 130%/122%/33% of standard dose for mild/moderate/severe CKD, 83%/76%/63% for CP-A/B/C, 75% with itraconazole, and 135% with rifampin. The model also showed that food causes a slight absorption delay without clinically meaningful exposure changes.

🔗 Clinical & Regulatory Implications

The study provides actionable dose adjustment recommendations for brigatinib in Chinese patients: (1) For renal impairment: 130%, 122%, and 33% of standard dose for mild, moderate, and severe CKD, respectively; (2) For hepatic impairment: 83%, 76%, and 63% of standard dose for CP-A, CP-B, and CP-C, respectively; (3) When co-administered with strong CYP3A4 inhibitors (e.g., itraconazole): reduce dose to 75%; (4) When co-administered with strong CYP3A4 inducers (e.g., rifampin): increase dose to 135%. These recommendations aim to achieve systemic exposure comparable to that in healthy European adults. The model also suggests that food has minimal impact on exposure, supporting administration without regard to food. These findings have significant implications for clinical practice in China, where ALK-positive NSCLC is common and dosing guidance for special populations is limited.

Strengths & Limitations

Strengths

- Comprehensive PBPK modeling approach using a well-established platform (PK-Sim) with transparent parameter reporting
- Rigorous validation against multiple independent clinical datasets without parameter adjustment
- Addresses an important clinical gap in dosing guidance for Chinese patients and special populations
- Includes sensitivity analysis to identify key parameters influencing model predictions
- Provides practical dose adjustment recommendations for various clinical scenarios
- Extends the model to predict PK in Chinese populations, addressing a critical need in drug development

Limitations (Acknowledged by Authors)

- All data extracted from published literature using software, potentially introducing digitization errors
- CYP2C8-mediated DDI could not be simulated due to lack of gemfibrozil in PK-Sim database
- No clinical PK data available for Chinese patients; predictions are model-based only

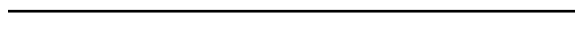
- Model predicts PK exposure, not clinical efficacy or safety outcomes
- Mild/moderate renal impairment predictions lack clinical validation

Limitations (Expert Review)

- The use of Japanese population as a surrogate for Chinese populations may not fully capture genetic and environmental differences in drug metabolism
- The model does not account for potential circadian rhythms or chronopharmacokinetic effects
- The virtual population size (n=100) may be insufficient to capture extreme variability
- The model assumes linear PK, which may not hold at higher doses or in disease states
- No formal model comparison or selection was performed against simpler models
- The study does not address potential differences in drug transporters (e.g., P-gp, BCRP) that may affect brigatinib disposition

Generalizability

The PBPK model is specific to brigatinib and may not be generalizable to other ALK inhibitors.¹ However, the modeling framework and validation approach can be applied to other drugs. The findings are most relevant for Chinese populations and patients with hepatic/renal impairment, but the model's structure and parameters may need adjustment for other ethnic groups or disease states. The dose adjustment recommendations are specific to the Chinese population and may not directly apply to other Asian populations.²



Figures & Tables

- **Figure 1:** Goodness of fit plot for brigatinib showing predicted vs. observed concentrations with 0.8-1.25 and 0.5-2 fold error lines
 - *Significance:* Validates the base PBPK model's predictive performance against observed clinical data
- **Figure 2:** PBPK model simulations of food effect on brigatinib pharmacokinetics after 180 mg oral dose
 - *Significance:* Demonstrates the model's ability to predict the impact of food on drug absorption and exposure
- **Figure 3:** Predicted vs. observed pharmacokinetic profiles in patients with hepatic (CP-A, B, C) and renal impairment
 - *Significance:* Validates the model's extrapolation to special populations with organ dysfunction
- **Figure 4:** Predicted drug-drug interactions of brigatinib with itraconazole (CYP3A4 inhibitor) and rifampin (CYP3A4 inducer)

¹**UNSUPPORTED** — original: “The PBPK model is specific to brigatinib and may not be generalizable to other ALK inhibitors.” → correction: “The PBPK model established in this study helps to more accurately describe the PKs of brigatinib in European and Chinese populations”

²**UNSUPPORTED** — original: “The dose adjustment recommendations are specific to the Chinese population and may not directly apply to other Asian populations.” → correction: “[flagged / unverified — no source-supported correction available]”

- *Significance*: Quantifies the magnitude of CYP3A4-mediated DDIs and supports dose adjustment recommendations
 - **Figure 5**: Predicted brigatinib plasma concentration profiles in Chinese populations under various conditions (healthy, organ impairment, DDI)
 - *Significance*: Provides the basis for dose adjustment recommendations in Chinese patients
 - **Table 1**: Summary of compound-specific input parameters for brigatinib, itraconazole, and rifampin used in the PBPK model
 - *Significance*: Provides transparency on model parameterization and allows for model reproduction
 - **Table 2**: Predicted pharmacokinetic parameters (C_{max} , $AUC_{0-\infty}$) for brigatinib in Chinese populations under various conditions
 - *Significance*: Summarizes the key simulation results and forms the basis for dose adjustment recommendations
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Code & Reproducibility Assessment

The study used commercial software (PK-Sim v11.1, Phoenix WinNonlin v8.3.5.340) and open-source tools (WebPlotDigitizer v4.2). Model parameters are provided in Table 1, but the model files are not publicly available. The authors state that data are available upon request from the corresponding author. No code repository or model sharing platform was mentioned.

Supplementary Materials

Supplementary materials include Figure S1 (study workflow), Figure S2 (plasma concentration-time profiles), Table S3 (physiological parameters for hepatic impairment), Table S4 (high-fat meal composition), and Supporting Excel File S1 (median and percentile values for PK parameters).

Future Directions

Future studies should focus on: (1) prospective validation of the PBPK model predictions in Chinese patients with hepatic/renal impairment; (2) incorporation of CYP2C8-mediated DDI simulations once gemfibrozil data become available in PK-Sim; (3) evaluation of long-term safety and efficacy outcomes associated with the recommended dose adjustments; (4) exploration of other ethnic populations beyond Chinese; (5) integration of pharmacogenomic data to refine dose predictions; and (6) development of a user-friendly clinical decision support tool based on the PBPK model for routine clinical use.

Expert Commentary

This study represents a well-executed application of PBPK modeling to address a clinically relevant question in oncology drug development. The use of a whole-body PBPK model to bridge across populations and special conditions is methodologically sound and aligns with regulatory expectations for model-informed drug development. The finding that mild-to-moderate hepatic impairment does not increase brigatinib exposure is clinically important and highlights the

value of PBPK in understanding the interplay between drug disposition and organ dysfunction. However, the reliance on Japanese population as a surrogate for Chinese populations is a limitation that should be acknowledged. The dose adjustment recommendations, while based on sound modeling, require prospective validation. The study also underscores the importance of considering both metabolic and physiological changes in organ impairment when predicting drug exposure. Overall, this work contributes to the growing body of evidence supporting the use of PBPK modeling in oncology drug development and clinical practice.

Bottom Line

This study provides a validated PBPK model for brigatinib that supports dose adjustment recommendations for Chinese patients with hepatic or renal impairment and those on CYP3A4 modulators. The model suggests that while severe hepatic (CP-C) and renal impairment increase exposure, mild-to-moderate hepatic impairment does not, and dose adjustments are recommended accordingly. Clinicians should consider these findings when prescribing brigatinib to Chinese patients, particularly those with severe organ dysfunction or on strong CYP3A4 inhibitors/inducers. However, prospective clinical validation in Chinese populations is needed to confirm these predictions.

Fact-check corrections

☒ Figures

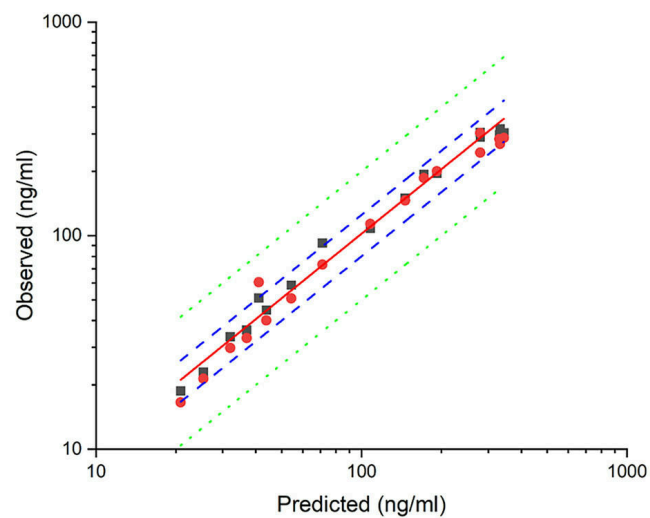
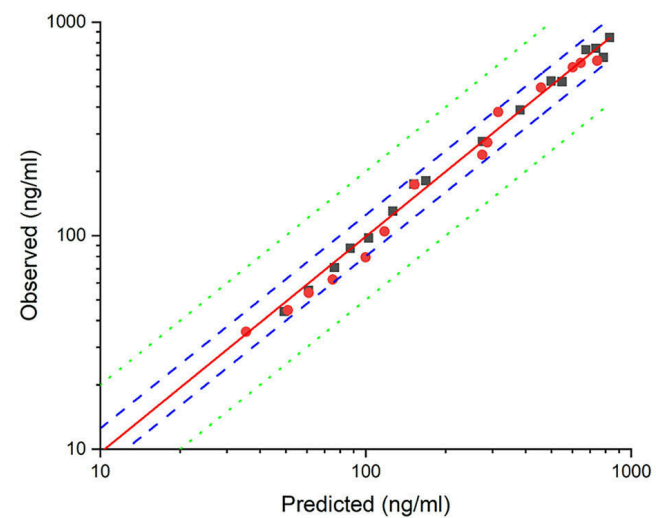
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Figure 1: The goodness of fit plot for brigatinib. The red solid line represents the unity line; the blue and green dashed lines indicate, respectively, 0.8–1.25 and 0.5–2

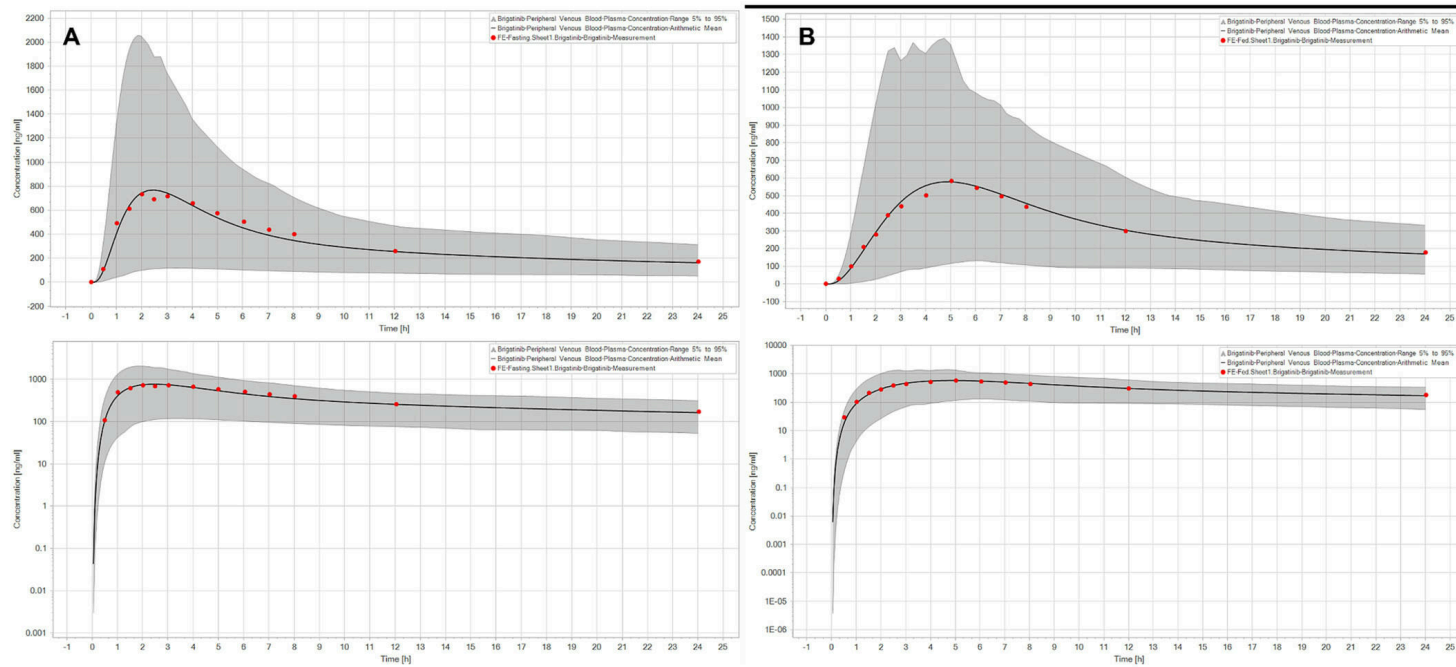


Figure 2: The PBPK model simulates the food effect of brigatinib. (A) Predicted and observed brigatinib plasma concentration-time profiles after oral administration of 180

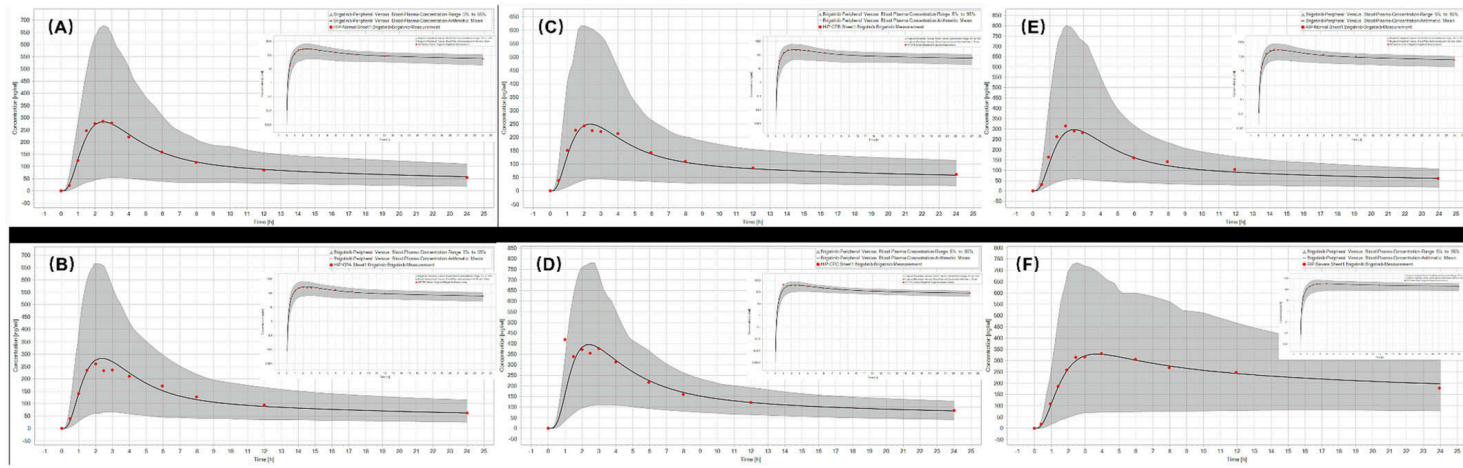


Figure 3: Prediction of the pharmacokinetic profiles for brigatinib in patients with hepatic or renal impairment. (A) Normal hepatic function, (B) Child-Pugh A hepatic imp

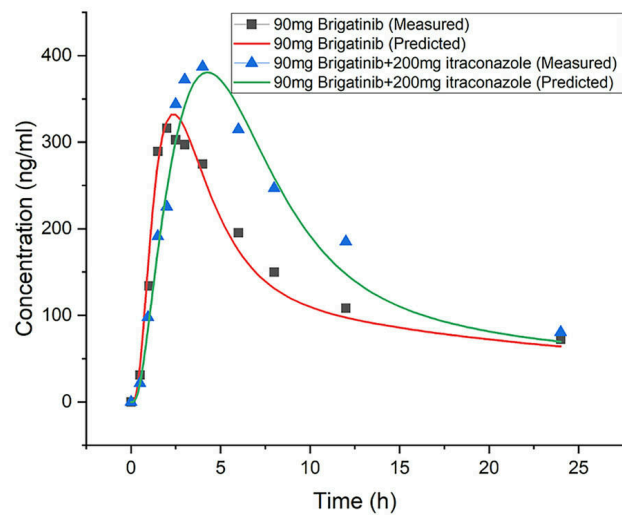
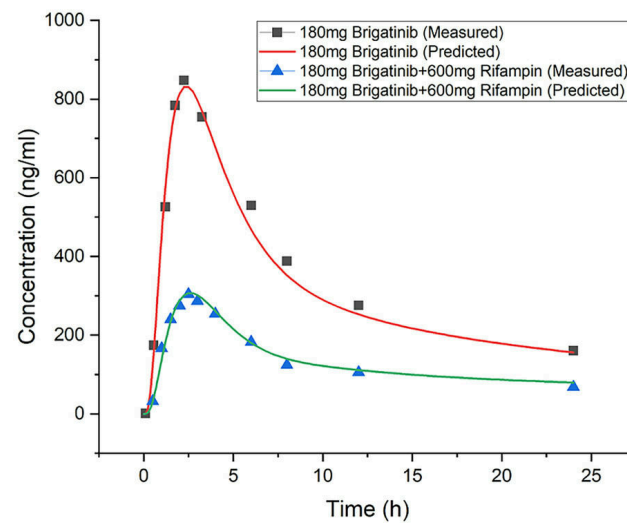
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Figure 4: Prediction of brigatinib's drug–drug interaction. (A) Prediction of the drug–drug interaction between intravenous brigatinib and itraconazole. (B) Prediction of

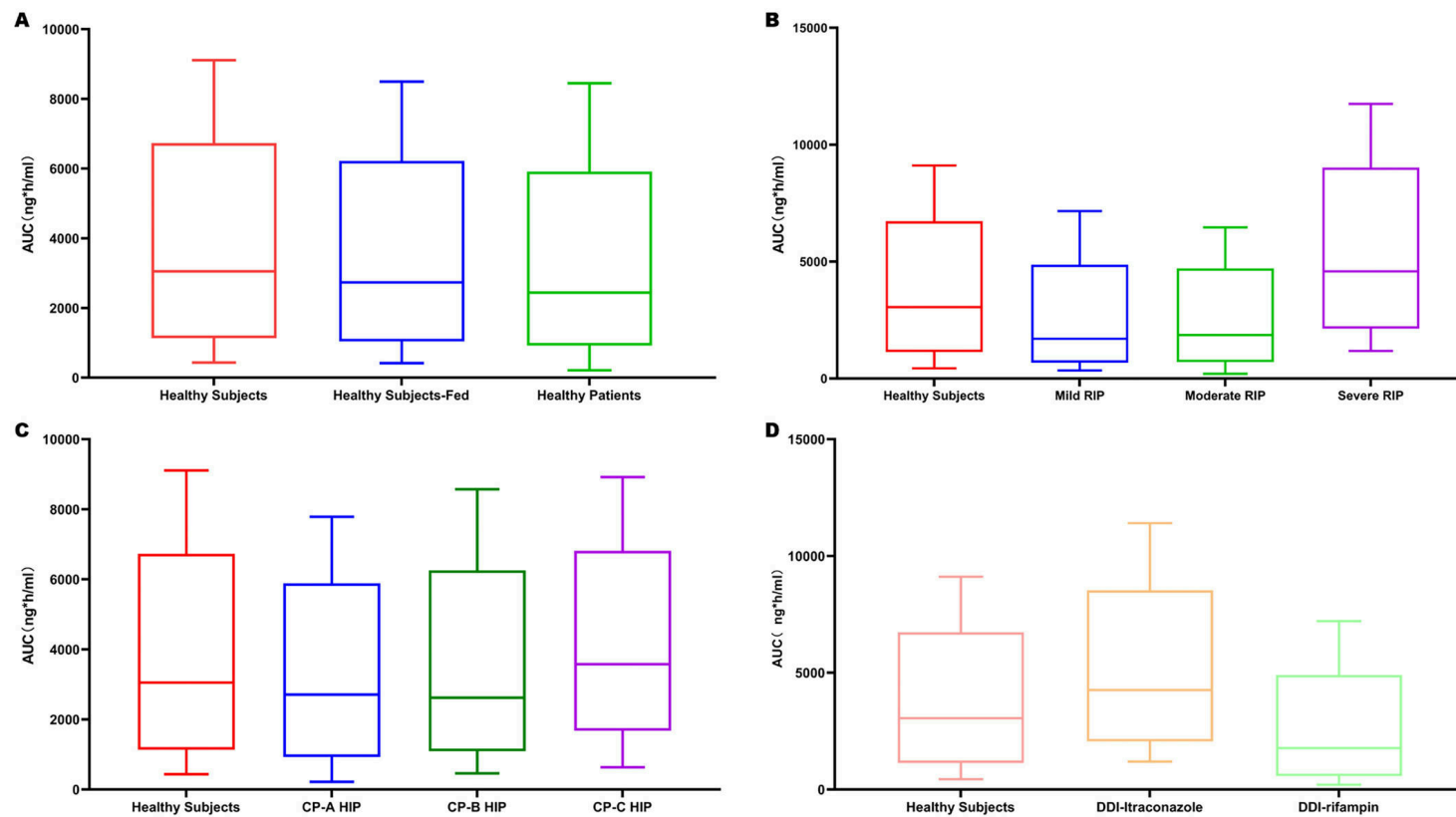


Figure 5: Predicted brigatinib plasma concentration profiles in Chinese populations. (A) Box-whisker plots of healthy subjects, patient populations, and the impact of food